

What Lets Data Answer a Causal Question?

Module C: Foundations · identification assumptions

August 5, 2026 · 12:20–1:05 ET

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The causal estimand is defined on outcomes we never fully see:

$$\Psi = E\left[Y^{(1)}\right] - E\left[Y^{(0)}\right]$$

- Each person contributes one outcome—under the exposure they actually had
- The other counterfactual is missing for everyone
- No sample size fixes this: it is a design problem, not a precision problem

Identification assumptions are the bridge

Three assumptions let an observed-data quantity stand in for the counterfactual contrast: **exchangeability**, **positivity**, and **consistency**.

$$Y^{(a)} \perp\!\!\!\perp A \mid W \quad \text{for } a \in \{0, 1\}$$

- Within strata of the measured covariates W , the exposed and unexposed are comparable—exposure is as good as random
- Pneumonia example: age and prior pneumonia drive both vaccination and pneumonia risk; conditioning on them restores the comparison
- The DAG is the argument: the adjustment set W must block every backdoor path

Not checkable from the data

No test can confirm exchangeability. It rests on the causal model, and its failure is quantified only by sensitivity analysis.

$$0 < P(A = 1 \mid W = w) < 1 \quad \text{for every } w \text{ with } P(W = w) > 0$$

- Every kind of person in the study could have received either exposure level
- **Structural violation:** a subgroup never receives one level—e.g. every long-term-care resident is vaccinated by policy
- **Random violation:** a covariate stratum is so sparse that one arm is empty by chance

Checkable from the data

Propensity-score overlap diagnostics make positivity visible—the TMLE walkthrough (Module G) inspects overlap directly.

$$A = a \Rightarrow Y = Y^{(a)}$$

- The observed outcome under exposure a equals the counterfactual outcome for that exposure
- Requires a well-defined exposure: “vaccinated” with which product, at what time, under what schedule?
- This is why the target-trial protocol pins down the treatment strategies before any analysis
- Companion condition (no interference): one person’s exposure does not change another person’s outcome

Vague interventions break the bridge

If the intervention is not specified well enough to imagine assigning it, the counterfactual—and the causal question—is not defined.

Together they identify the causal effect from observable data:

$$E\left[Y^{(a)}\right] = \sum_w E[Y \mid A = a, W = w] P(W = w)$$

Assumption	In words	Checkable?	Defended by
Exchangeability	no unmeasured confounding	no	DAG + sensitivity analysis
Positivity	both exposures occur everywhere	yes	overlap diagnostics
Consistency	well-defined intervention	no	target-trial protocol

- The Roadmap (Module D) is where each assumption is declared, defended, and revisited
- No estimator—however modern—can rescue a failed assumption: TMLE estimates the identified quantity, nothing more